



Lessons 1 and 2: What to Keep

Technologies for Artificial Intelligence

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Lesson 1: what to keep

- We minimise the error on the sample but care about new data: so hold data out
- Baseline first: always “benign” scores **62.9%**
- With 1% positives: **0.986** accuracy, **0** recall
- **77%** on coin-flip labels, from choosing features before the split
- Four traps raise no error: leakage, imbalance, shortcuts, one split

Lesson 2: what to keep

- Look before you touch: types, gaps, shapes
- Ask **why** a value is missing, not how many
- A tenure of **-3** passes both outlier rules
- Every encoding makes a claim about categories
- **94 of 128** imputed rows borrowed from the test set
- What learns from data is fitted on training rows only